

Lab 5: Biofuels

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Abstract

This lab was conducted in order to explore a way to make an effective bio fuel, as well as to measure the heat of combustion of several fuels. A side goal of the lab was to gain proficiency with using a bomb calorimeter. Since combustion engines are incredibly prevalent in our world, understanding how to measure various properties of common fuels is very useful. The first step of this lab was to create a bio fuel using commonly available materials. Then, a bomb calorimeter was calibrated using a fuel with a known heat of combustion in order to find the effective heat capacity of the calorimeter. Finally, the heats of combustion of four fuels, including the bio fuel, were measured and analyzed. It was determined that the heat of combustion of the bio fuel that was created was 9,568.3 calories/gram. It was also determined that the heats of combustion of diesel, carbon, and a common granola bar were 13,666.4 cal/g, 6,266.8 cal/g, and 5,992.6 cal/g respectively. Therefore, it was found that while bio fuel is fairly effective, it is not quite as efficient as traditional diesel. Additionally, it was found that in all of the tests, the mixtures were very fuel lean.

1 Introduction

The purpose of this lab was primarily to explore the heat of combustion of various fuels. A secondary purpose was to make a bio fuel and also test its properties. One reason for researching this topic is to gain proficiency with measuring various properties of fuels. Since there are still many engineering applications that use engines rather than electric motors, being able to analyze the properties of different fuels is quite useful. For example, biodiesel, natural gas, ethanol, and propane are all viable fuels with different properties that might be used in different applications^[1]. Additionally, not only are there a lot of different types of fuels, there are lot of combustion engines in use all around the world. According to one source, there are around "1.4 billion internal combustion engine vehicles worldwide today," (Lyseng)^[3]. Therefore, there is clearly a great need to keep improving combustion engines, which could also mean improving and working with different fuels. Another reason to research this topic is that the heat of combustion of a fuel is often very strongly related to the efficiency of an engine^[2]. Therefore, if one wants to build a very efficient engine, it is important to be able to understand the heat of combustion properties of various potential fuels. In order to test the properties of various fuels, a bomb calorimeter was used in conjunction with a thermistor in order to record the temperature change over time of the system. First a fuel with a known heat of combustion was tested in order to determine the effective heat capacity of the bomb calorimeter, and then four other fuels with unknown heats of combustion were tested. After the heats of combustion of the four fuels were determined, the results were analyzed, in particular the results of the bio fuel in comparison to the other fuels. The bio fuel was made in order to explore the process with which an effective fuel can be made.

2 Experimental Methods

The lab began with reading over the provided lab manual in order to get a sense for the direction and purpose of the experiments. Then, it was determined that the first step was to create Bio fuel that was later used as a fuel in the heat of combustion experiments. This had to be the first step since a step in the process of creating the Bio fuel was to let the fuel separate over the course of a day or more. Therefore, the Bio fuel was made on a Friday, and then the rest of the lab was conducted on the following Thursday. In order to

create the Bio fuel, Canola oil, Methanol, and Potassium Hydroxide were combined. The first step was to measure out 500 mL of Canola oil and set it aside in a beaker. Then, 100 mL of Methanol was measured out and set aside in a second beaker. After that, 2.52 grams of Potassium Hydroxide was measured out using a scale and then was placed into the 100 mL of Methanol. After the Potassium Hydroxide was placed in the Methanol, the mixture was swirled until the Potassium Hydroxide (which was initially in solid pellet form) was fully dissolved in the Methanol. This created a mixture of Methoxide in one beaker. It is important to note that during the swirling of the Methanol and Potassium Hydroxide beaker, some of the solution splashed onto the researchers hand that was swirling the solution. The amount of solution that splashed out was very small, so the procedure proceeded like normal after the researcher washed their hand, but it is still important to note as this could have been a potential source of error in the final results. After the Methoxide solution was created, the solution was carefully poured into a blender. Then, the Canola oil was poured into the same blender, with care taken to not cause any of the solution to splash out. After the the Canola oil and the Methoxide were both in the blender, the lid was placed on the blender and the blender was run for a few minutes in order to insure complete mixing of the final solution. The blender setup was as shown below in Figure 1:



Figure 1: Bio Fuel Blending

After several minutes of blending, the blender was stopped and some of the mixed solution was poured into a mason jar. The mason jar was then closed and the names of the researchers was written on the top of the jar in order to prevent any confusion with other lab groups. Then, the lab space was cleaned up, the excess solution that wasn't used was disposed of, and the solution was left to settle over the course of several days. It is important to note that throughout the creation of the Bio fuel, much care was taken in order to ensure the safety of the researchers. Since Methanol is quite dangerous, gloves and safety goggles were worn throughout the Bio fuel creation.

After several days passed and the Bio fuel had fully settled, the second half of the lab could be conducted. The second half of the lab was aimed at determining the heat of combustion of various fuels, including the Bio fuel. This was done using a bomb calorimeter in conjunction with LabVIEW. Five total fuels were tested: Benzoic acid (which was the control), Bio fuel, Diesel, Carbon, and a bit of a granola bar. The first fuel that was tested was the Benzoic acid (C_6H_5COOH) since the heat of combustion of Benzoic acid at a constant volume was known before the tests were conducted. This allowed the Benzoic acid test to be used as a calibration of the system. Since the effective heat capacity of the calorimeter was not known before hand, it had to be determined by experimentation. The Benzoic acid test was used to determine the effective heat capacity of the calorimeter. The Benzoic acid test was conducted as follows.

First, around 1 gram of Benzoic acid was measured out using a scale (1.017 grams to be exact). Then, the acid was placed into the fuel holder in the bomb. Next, 10 cm of ignition wire was measured out and placed in the bomb ignition place. It was formed so that as much of the wire as possible was directly over the fuel holder to maximize the chance of a successful ignition. Then, the top of the bomb (which had the fuel holder and ignition place attached to it) was connected and screwed onto the main casing of the bomb.

The bomb setup is shown below in Figure 2:

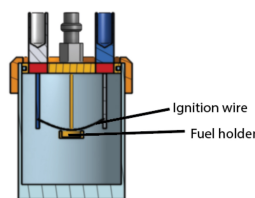


Figure 2: Bomb Cross-Section

The bomb was then pressurized using pure oxygen. During the pressurization, the air was purged from the bomb in order to ensure that only pure oxygen was left. All of the tests were pressurized to about 20 atm with oxygen. The top of the bomb with its pressurization valves and ignition connections is shown below in Figure 3:

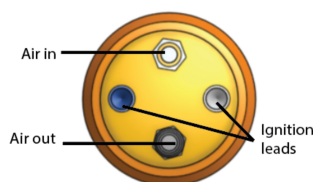


Figure 3: Top of Bomb

After the bomb was filled with the fuel and the pressurized oxygen, it was placed into a quasi-adiabatic calorimeter. After the bomb was placed in the calorimeter, wires that were integrated with the calorimeter were connected to the ignition leads at the top of the bomb. On the outside of the calorimeter, there was a button that when pressed would send a current through the ignition wire and therefore ignite the bomb. After the wires were connected, the space in between the bomb and the walls of the calorimeter was filled with 2 L of water which was measured out before hand. This was done in order to surround the bomb with a buffer that had a very high specific heat. Since the water has a very high specific heat, it prevented the temperature change from being too large. Additionally, the water created a buffer whose temperature would be very easy to measure. After the water was poured into the calorimeter (which completely surrounded the bomb in water), the lid of the calorimeter was attached. The lid contained two important pieces of equipment: a thermometer and a mixer. The thermometer (which was a precision 10 k Ω thermistor) was long in order to allow it to be submerged in the water and therefore gain an accurate temperature measurement of the water. The mixer was there in order to constantly cycle the water around the calorimeter so that no temperature gradients were built up in the water. This ensured that the temperature of the water was constant throughout the calorimeter and that the temperature reading from the stationary thermometer would be accurate. The full bomb calorimeter setup is shown below in Figure 4:

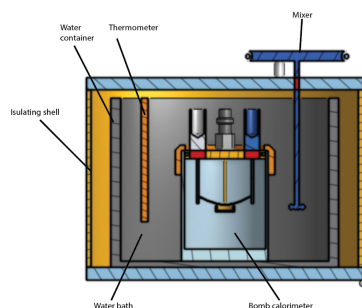


Figure 4: Bomb Calorimeter Experimental Setup

Additionally, the thermometer had to be wired into a breadboard according to the circuit shown below in Figure 5:

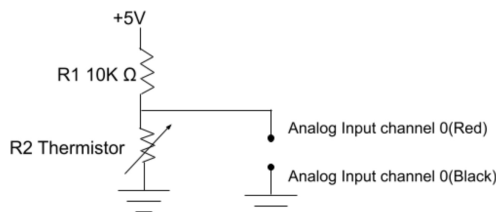


Figure 5: Thermistor Circuit

The output of this circuit was connected to a DAQ board which was then connected to a computer. This was done in order to allow the data to be collected and recorded by LabVIEW. After the setup was complete, the first test was ready to be conducted.

The test began by first turning on the mixer. Then, shortly after turning on the mixer, the LabVIEW program used to collect the temperature versus time data of the system was started. The LabVIEW program was run with the mixer going for about 30 seconds before the bomb was ignited. This allowed for data about the initial state of the system to be collected. After the ignition, the system was left untouched and the LabVIEW program was left running for about 9-10 minutes. This time allowed for the water to get to its maximum temperature. After it appeared on the LabVIEW program that the water was just beginning to cool down, the program was stopped. The mixer was then stopped and the calorimeter lid was removed. The ignition wires were disconnected, and the bomb was removed from the calorimeter. The pressure was slowly released from the bomb so that the bomb could be opened up. After the bomb was removed, two key steps happened to finalize the first test. First, the bomb was opened up, and all of the excess, unburned ignition wire was collected and measured. This was to account for any error that might have been introduced due to incomplete burning of the ignition wire. The second was that the water was drained from the calorimeter. In fact, for each of the five tests, different, fresh water was used. This was done to ensure that the heat capacity of the water didn't vary from test to test. After the water was drained and the excess wire was accounted for, the first test was done.

After the first test, the same procedure was repeated, but for four other fuels. About 1 gram of each fuel was measured out for each of the other tests. For each of the tests, the amount of fuel was recorded, as well as the exact pressure of the bomb, the amount of excess ignition wire, and the temperature versus time data. After all of the tests were conducted, the experimental setup was cleaned up, and all of the data was saved for later analysis.

3 Results

For each of the five tests, the mass of the fuel, the amount of excess ignition wire, and the pressure of the bomb was recorded. Those results/measurements, which aren't important in and of themselves but were used for later analysis, are recorded below in the table in Figure 6:

	Benzoic Acid	Bio Fuel	Diesel	Carbon	Granola Bar
Mass of Fuel (grams)	1.017	1.007	1.004	1.003	1.015
Pressure of Bomb (atm)	20.30	21.10	20.82	20.50	20.40
Excess Wire (calories)	15.5	12.0	15.0	4.0	13.0

Figure 6: Important Measurements from the Tests

Using these values, the tests can be analyzed and the effective heat capacity of the bomb calorimeter as well as the heat of combustion of the fuels can be determined.

Now looking at the first test with the Benzoic acid, the data can be used to determine the effective heat capacity of the bomb calorimeter. The equation used to find this is given below:

$$C = \frac{q_f m_f + l_{wire} q_{wire}}{\Delta T}$$

Where C is the effective heat capacity of the bomb calorimeter in $\left[\frac{\text{cal}}{\text{K}}\right]$, q_f is the heat of combustion of the fuel (Benzoic acid) per unit mass at a constant volume in oxygen $\left[\frac{\text{cal}}{\text{g}}\right]$, m_f is the mass of the fuel [g], l_{wire} is the length of combusted wire [cm], q_{wire} is the heat of combustion per unit centimeter of the ignition wire $\left[\frac{\text{cal}}{\text{cm}}\right]$ which is known to be 2.3, and ΔT is the change of temperature before and after the combustion [K]. In this equation, q_f is already known to be 6,318 $\frac{\text{cal}}{\text{g}}$ for Benzoic acid. Also in the equation, $l_{wire} q_{wire}$ as well as m_f are known from the table in Figure 6. That just leaves ΔT to be determined in order to find C , which can be found using the data from the first test. The data from the Benzoic acid test are shown below in Figure 7:

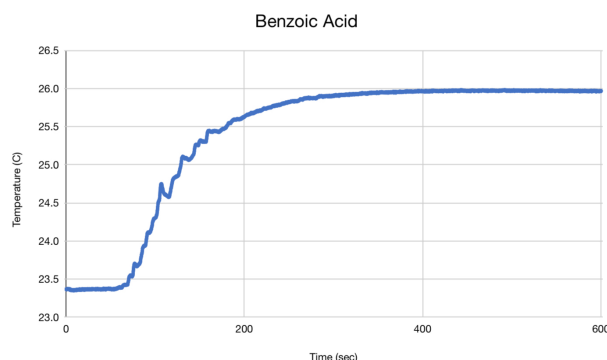


Figure 7: Temperature vs. Time Data for Benzoic Acid Test

From this data, it can be seen that the system begins at a low initial temperature, and then after the bomb is ignited, slowly climbs in temperature until it reaches a maximum final temperature. In order to find the initial temperature, the first 10 seconds of data was used and the average temperature over those first 10 seconds was determined. It is important to note that the LabVIEW program took a data point every 0.1 seconds, so 10 seconds of data comes out to 100 total data points. A similar procedure was used to determine the final temperature for the test. To determine the final temperature, the last minute of data was averaged to get one average temperature measurement. Since the calorimeter was quasi-adiabatic, it is assumed for the analysis that the calorimeter was fully adiabatic, so it was assumed that after the system reached its peak temperature, essentially no heat was lost, so taking the last minute of data represents the final peak temperature. Additionally, the standard deviation of the temperature values over the first time period was taken in order to get an error estimate of the temperature variation. Using these methods, the Benzoic acid temperature data and error is as follows: initial temperature: 23.360°C, final temperature: 25.968°C, temperature error: $\pm 0.007^\circ\text{C}$. Therefore, $\Delta T = 2.608 \text{ K}$ (since $\text{K} = 273.15 + \text{C}$, the ΔT is the same) with an error of $\pm 0.007 \text{ K}$. Therefore, the effective heat capacity of the bomb calorimeter can be calculated as follows:

$$C = \frac{(6,318)(1.017) + ((10)(2.3) - (15.5))}{(2.608)} = 2,466.6 \left[\frac{\text{cal}}{\text{K}}\right]$$

Using this result in tandem with the rest of the experimental data, the heat of combustion of the other four fuels can be determined. However, before that happens, the error of the effective heat capacity needs to be determined. It can be assumed that there is no error in q_{Benzoic} or in q_{wire} since those were given values. The error in the mass can be found by estimating the uncertainty in the precision of the scale that was used. In this case, the scale displayed three decimal points, so it can be assumed that the scale error, and in turn the mass error, is ± 0.0005 grams. The error in the length of the wire can be found using the uncertainty in

the precision of the ruler used to measure out the wire, which comes out to be an error of ± 0.2 cm. And finally, the last variable in this equation with error associated with it is the ΔT , which was already found to be an error of ± 0.007 K. These errors can then be combined and propagated in order to find the error in the value for C . The error propagation equation is as follows:

$$C_{error} = \sqrt{\left(\frac{q_f}{\Delta T} m_{f,error}\right)^2 + \left(\frac{q_{wire}}{\Delta T} l_{wire,error}\right)^2 + \left(\frac{q_f m_f + l_{wire} q_{wire}}{\Delta T^2} T_{error}\right)^2}$$

After plugging in all of the values above into this equation, it can be found that $C_{error} = 6.7 \left[\frac{\text{cal}}{\text{K}}\right]$. Now, with the effective heat capacity of the bomb calorimeter and its associated error, the rest of the tests can be analyzed.

For the four fuels that have unknown heats of combustion (namely the bio fuel, carbon, diesel, and granola bar), the following equation can be used to find their heats of combustion:

$$q_f = \frac{C\Delta T - l_{wire}q_{wire}}{m_f}$$

All of the variables are the same as in the equation used to find the effective heat capacity C of the bomb calorimeter. The difference is that the equation has been rearranged to solve for q_f rather than for C . Additionally, the error present is the same as was present in the analysis of the Benzoic acid test, except that now there is the additional error of the effective heat capacity, which was determined above. Also, the error in the temperature change will be slightly different for each of the following four tests since the temperature error (like the Benzoic acid test) will be found by calculating the standard deviation of the first 100 data points of each of the tests. And, the error propagation equation changes slightly for the analysis of the following four tests. The error propagation equation is as follows:

$$q_{f,error} = \sqrt{\left(\frac{C}{m_f} T_{error}\right)^2 + \left(\frac{-q_{wire}}{m_f} l_{wire,error}\right)^2 + \left(\frac{-(C\Delta T - l_{wire}q_{wire})}{m_f^2} m_{f,error}\right)^2 + \left(\frac{\Delta T}{m_f} C_{error}\right)^2}$$

So, with the above equations for q_f and $q_{f,error}$, as well as with all of the known and calculated values, the only values left to analyze the following four tests are the change in temperature for each test and the temperature error for each test. Just as with the Benzoic acid test, the initial temperature will be found by averaging the first 100 data points, the final temperature will be found by averaging the last 600 data points, and the temperature error will be found by finding the standard deviation in the first 100 data points. First, the bio fuel test will be analyzed. The raw data for the bio fuel test is shown below in Figure 8:

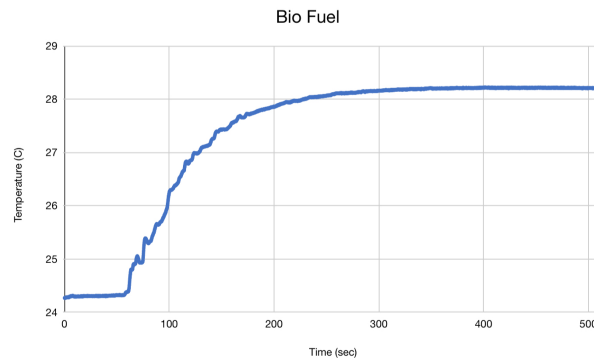


Figure 8: Temperature vs. Time Data for Bio Fuel Test

Using the methods used to analyze the Benzoic acid test, the initial temperature, final temperature, and temperature error for the bio fuel test can be found. They are as follows: initial temperature: 24.288°C , final

temperature: 28.211°C, temperature error: $\pm 0.013^\circ\text{C}$. Therefore, $\Delta T_{bio} = 3.923\text{ K}$. Plugging these results into the above equations, it can be found that:

$$q_{bio} = 9,598.3 \pm 41.5 \left[\frac{\text{cal}}{\text{g}} \right]$$

Next, the diesel test will be analyzed. The raw data for the diesel test is shown below in Figure 9:

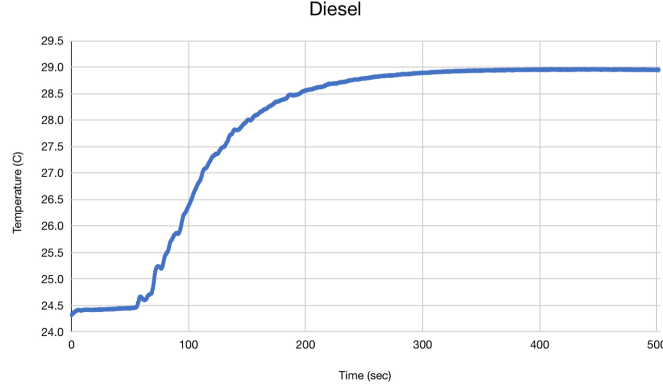


Figure 9: Temperature vs. Time Data for Diesel Test

For the diesel test, the temperature data is as follows: initial temperature: 23.387°C, final temperature: 28.953°C, temperature error: $\pm 0.028^\circ\text{C}$. Therefore, $\Delta T_{diesel} = 5.566\text{ K}$. Therefore, it can be found using the above equations that:

$$q_{diesel} = 13,666.4 \pm 78.5 \left[\frac{\text{cal}}{\text{g}} \right]$$

Next, the carbon test will be analyzed. The raw data for the carbon test is shown below in Figure 10:

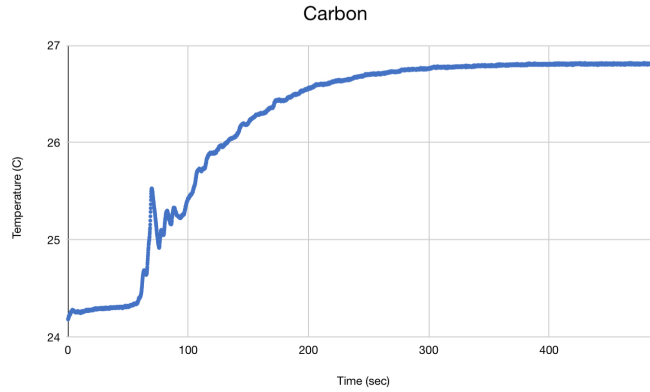


Figure 10: Temperature vs. Time Data for Carbon Test

For the carbon test, the temperature data is as follows: initial temperature: 24.253°C, final temperature: 26.809°C, temperature error: $\pm 0.021^\circ\text{C}$. Therefore, $\Delta T_{carbon} = 2.556\text{ K}$. Therefore, it can be found using the above equations that:

$$q_{carbon} = 6,266.8 \pm 54.5 \left[\frac{\text{cal}}{\text{g}} \right]$$

And finally, the granola bar test will be analyzed. The raw data for the granola bar test is shown below in Figure 11:

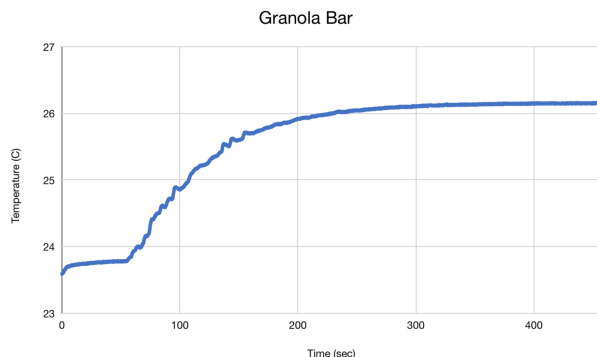


Figure 11: Temperature vs. Time Data for Granola Bar Test

For the granola bar test, the temperature data is as follows: initial temperature: 23.678°C, final temperature: 26.148°C, temperature error: $\pm 0.041^\circ\text{C}$. Therefore, $\Delta T_{\text{granola}} = 2.470\text{K}$. Therefore, it can be found using the above equations that:

$$q_{\text{granola}} = 5,992.6 \pm 101.0 \left[\frac{\text{cal}}{\text{g}} \right]$$

All of these results are summarized in the table below in Figure 12:

	Benzoic Acid	Bio Fuel	Diesel	Carbon	Granola Bar
Heat of Combustion (cal/g)	6,318.0	9,598.3	13,666.4	6,266.8	5,992.6
Error (cal/g)	0	41.5	78.5	54.5	101.0

Figure 12: Heat of Combustion Data for All Tests

The last part of the analysis of the various fuels is to determine the ratio of the fuels to the oxygen in their respective tests, as well as to determine if the fuel mixtures were rich or lean. This was done by finding the fuel air ratio and the mixture equivalence ratio for all of the fuels (except for the granola bar since its chemical composition was unknown). First, the mass of oxygen in each test was calculated. The mass of oxygen can be calculated using the following equation:

$$m_{\text{oxygen}} = (\text{Molar mass})(n)$$

Where the molar mass of oxygen is 31.999, and n is the number of moles of oxygen. In order to find n, the following equation can be used:

$$n = \frac{PV}{ZRT}$$

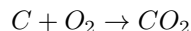
Where P is the pressure, V is the volume, Z is the compressibility factor of oxygen at a given pressure, R is the universal gas constant, and T is the temperature. The only two variables that changed from test to test were the pressure and temperature, and they were very slight changes. Therefore, the moles were found individually for each test, but only one example calculation is shown below for the bio fuel test:

$$n = \frac{(21.1)(0.35)}{(0.86)(0.0821)(297.41)} = 0.352$$

Therefore, the mass of the oxygen in the bio fuel test was:

$$m_{oxygen} = 11.25 \text{ grams}$$

With this process, the fuel to air ratios can be found for all of the tests. In order to find the mixture equivalence ratio to determine if the mixtures were fuel rich or fuel lean, the stoichiometric fuel to air ratios need to be found. In order to find this stoichiometric ratio, the chemical reaction equation for each test needs to be found, and then the mass ratio between the fuel and the oxygen can be found. For example, the carbon-oxygen combustion equation is as follows:



Since the molar mass of carbon is 12.011, and one mole of carbon is used, and since the molar mass of O_2 is 31.999, and one mole of O_2 is used, the stoichiometric fuel to air ratio can be found as follows:

$$FAR_{stoich} = \frac{12.011}{31.999} = 0.375$$

Using the above procedures, the fuel to air ratios of all of the tests was determined, as well as the mixture equivalence ratios for all of the tests. The data is shown below in the table in Figure 13:

	Fuel Air Ratio	Mixture Equivalence Ratio
Biodiesel	0.2946	0.3037
Diesel	0.2946	0.3070
Carbon	0.3754	0.2443
Benzoic Acid	0.5088	0.1840

Figure 13: FAR and MER for All Tests

4 Discussion

The results of the first part of the experiment, the Benzoic acid test, weren't surprising or particularly useful in and of themselves, but they were instead used to analyze the rest of the tests. Therefore, the discussion will begin with the bio fuel test. As can be seen in Figure 8, the bio fuel test demonstrates a sharp and sizable jump in the temperature of the system over time. With a change in temperature of 3.923 K, and a heat of combustion of 9,598.3 cal/g, it had the second highest change in temperature of any fuel, and the second highest heat of combustion of any fuel. It was second only to traditional diesel. This is quite impressive given that the bio fuel was created in the lab less than a week before it was tested, and it was created with easily obtainable materials. Next, looking at the diesel it is clear that it has the greatest change in temperature and the highest heat of combustion. In fact its change in temperature is 5.566 K while the second greatest (bio fuel) is 1.643 K less. Additionally, its heat of combustion is 13,666.4 cal/g, which is 4,068.1 greater than the next best (also bio fuel). It also has over double the heat of combustion of both carbon and the granola bar. Looking at carbon, it is very middle of the pack in terms of its change in temperature and its heat of combustion. Interestingly, in Figure 10, there is a very noticeable spike in its temperature vs. time data while the system was heating up. This was likely due to uneven mixing of the water, but it is still very interesting to see. And finally, looking at the granola bar, it has the lowest change in temperature and the lowest heat of combustion of any of the fuels. This is understandable since it is consumable food, and it is actually surprising how well it did. Its change in temperature was only 0.086 K off from the carbon.

Now looking at the results that were found in this lab for the heat of combustion for the four well known fuels (not the granola bar), this lab's results can be compared to commonly agreed upon heats of combustion for these fuels. First looking at diesel, one source says that the heat of combustion of standard diesel ranges from 10,038.2 to 10,994.3 cal/g typically^[4]. Comparing that with this experiment's value, it can be seen that the value found in this lab of 13,666.4 cal/g is significantly higher. This is potentially due to varying ignition environments, since in this lab it was specifically pressurized to around 20 atm with pure oxygen. Other than that, it is unclear what exactly could have caused this error because all of the known errors,

